

9 Register description

The device contains a set of registers which are used to control its behavior and to retrieve linear acceleration, angular rate and temperature data. The register addresses, made up of 7 bits, are used to identify them and to write the data through the serial interface.

9.1 FUNC_CFG_ACCESS (01h)

Enable embedded functions register (r/w).

Table 20. FUNC_CFG_ACCESS register

FUNC_CFG_EN	0 ⁽¹⁾	FUNC_CFG_EN_B	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾
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1. This bit must be set to '0' for the correct operation of the device.

Table 21. FUNC_CFG_ACCESS register description

FUNC_CFG_EN	Enable access to the embedded functions configuration registers bank A and B ⁽¹⁾ . Default value: 0. Refer to Table 22 .
FUNC_CFG_EN_B	Enable access to the embedded functions configuration register bank B ⁽¹⁾ . Default value: 0. Refer to Table 22 .

1. The embedded functions configuration registers details are available in [Section 10: Embedded functions register mapping](#), [Section 11: Embedded functions registers description - Bank A](#), and [Section 12: Embedded functions registers description - Bank B](#).

Table 22. Configuration of embedded functions register banks

FUNC_CFG_EN	FUNC_CFG_EN_B	Status of embedded register banks
0	0	Bank A and B disabled (default)
0	1	Forbidden
1	0	Bank A enabled
1	1	Bank B enabled

9.2 SENSOR_SYNC_TIME_FRAME (04h)

Sensor synchronization time frame register (r/w).

Table 23. SENSOR_SYNC_TIME_FRAME register

0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	TPH_3	TPH_2	TPH_1	TPH_0
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1. This bit must be set to '0' for the correct operation of the device.

Table 24. SENSOR_SYNC_TIME_FRAME register description

TPH_ [3:0]	Sensor synchronization time frame with the step of 500 ms and full range of 5 s. Unsigned 8-bit. Default value: 0000 0000 (sensor sync disabled)
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9.3 SENSOR_SYNC_RES_RATIO (05h)

Sensor synchronization resolution ratio (r/w)

Table 25. SENSOR_SYNC_RES_RATIO register

0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	RR_1	RR_0
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1. This bit must be set to '0' for the correct operation of the device.

Table 26. SENSOR_SYNC_RES_RATIO register description

RR_[1:0]	Resolution ratio of error code for sensor synchronization:
	00: SensorSync, Res_Ratio = 2-11
	01: SensorSync, Res_Ratio = 2-12
	10: SensorSync, Res_Ratio = 2-13
	11: SensorSync, Res_Ratio = 2-14

9.4 FIFO_CTRL1 (06h)

FIFO control register (r/w).

Table 27. FIFO_CTRL1 register

FTH_7	FTH_6	FTH_5	FTH_4	FTH_3	FTH_2	FTH_1	FTH_0
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Table 28. FIFO_CTRL1 register description

FTH_[7:0]	FIFO threshold level setting ⁽¹⁾ . Default value: 0000 0000.
	Watermark flag rises when the number of bytes written to FIFO after the next write is greater than or equal to the threshold level.
	Minimum resolution for the FIFO is 1 LSB = 2 bytes (1 word) in FIFO

1. For a complete watermark threshold configuration, consider FTH_[10:8] in [FIFO_CTRL2 \(07h\)](#).

9.5 FIFO_CTRL2 (07h)

FIFO control register (r/w).

Table 29. FIFO_CTRL2 register

TIMER_PEDO_FIF0_EN	TIMER_PEDO_FIF0_DRDY	0 ⁽¹⁾	0 ⁽¹⁾	FIF0_TEMP_EN	FTH10	FTH_9	FTH_8
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1. This bit must be set to '0' for the correct operation of the device.

Table 30. FIFO_CTRL2 register description

TIMER_PEDO_FIF0_EN	Enable pedometer step counter and timestamp as 4 th FIFO data set. Default: 0 (0: disable step counter and timestamp data as 4 th FIFO data set; 1: enable step counter and timestamp data as 4 th FIFO data set)
TIMER_PEDO_FIF0_DRDY	FIFO write mode ⁽¹⁾ . Default: 0 (0: enable write in FIFO based on XL/Gyro data-ready; 1: enable write in FIFO at every step detected by step counter.)
FIF0_TEMP_EN	Enable the temperature data storage in FIFO. Default: 0. (0: temperature not included in FIFO; 1: temperature included in FIFO)
FTH_[10:8]	FIFO threshold level setting ⁽²⁾ . Default value: 0000 Watermark flag rises when the number of bytes written to FIFO after the next write is greater than or equal to the threshold level. Minimum resolution for the FIFO is 1LSB = 2 bytes (1 word) in FIFO

1. This bit is effective if the DATA_VALID_SEL_FIF0 bit of the [MASTER_CONFIG \(1Ah\)](#) register is set to 0.
2. For a complete watermark threshold configuration, consider FTH_[7:0] in [FIF0_CTRL1 \(06h\)](#).

9.6 FIFO_CTRL3 (08h)

FIFO control register (r/w).

Table 31. FIFO_CTRL3 register

0 ⁽¹⁾	0 ⁽¹⁾	DEC_FIFO_GYRO2	DEC_FIFO_GYRO1	DEC_FIFO_GYRO0	DEC_FIFO_XL2	DEC_FIFO_XL1	DEC_FIFO_XL0
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1. This bit must be set to '0' for the correct operation of the device.

Table 32. FIFO_CTRL3 register description

DEC_FIFO_GYRO [2:0]	Gyro FIFO (first data set) decimation setting. Default: 000 For the configuration setting, refer to Table 33 .
DEC_FIFO_XL [2:0]	Accelerometer FIFO (second data set) decimation setting. Default: 000 For the configuration setting, refer to Table 34 .

Table 33. Gyro FIFO decimation setting

DEC_FIFO_GYRO [2:0]	Configuration
000	Gyro sensor not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

Table 34. Accelerometer FIFO decimation setting

DEC_FIFO_XL [2:0]	Configuration
000	Accelerometer sensor not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

9.7 FIFO_CTRL4 (09h)

FIFO control register (r/w).

Table 35. FIFO_CTRL4 register

STOP_ON_FTH	ONLY_HIGH_DATA	DEC_DS4_FIFO2	DEC_DS4_FIFO1	DEC_DS4_FIFO0	DEC_DS3_FIFO2	DEC_DS3_FIFO1	DEC_DS3_FIFO0
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Table 36. FIFO_CTRL4 register description

STOP_ON_FTH	Enable FIFO threshold level use. Default value: 0. (0: FIFO depth is not limited; 1: FIFO depth is limited to threshold level)
ONLY_HIGH_DATA	8-bit data storage in FIFO. Default: 0 (0: disable MSByte only memorization in FIFO for XL and Gyro; 1: enable MSByte only memorization in FIFO for XL and Gyro in FIFO)
DEC_DS4_FIFO[2:0]	Fourth FIFO data set decimation setting. Default: 000 For the configuration setting, refer to Table 37 .
DEC_DS3_FIFO[2:0]	Third FIFO data set decimation setting. Default: 000 For the configuration setting, refer to Table 38 .

Table 37. Fourth FIFO data set decimation setting

DEC_DS4_FIFO[2:0]	Configuration
000	Fourth FIFO data set not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

Table 38. Third FIFO data set decimation setting

DEC_DS3_FIFO[2:0]	Configuration
000	Third FIFO data set not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

9.8 FIFO_CTRL5 (0Ah)

FIFO control register (r/w).

Table 39. FIFO_CTRL5 register

0 ⁽¹⁾	ODR_FIFO_3	ODR_FIFO_2	ODR_FIFO_1	ODR_FIFO_0	FIFO_MODE_2	FIFO_MODE_1	FIFO_MODE_0
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1. This bit must be set to '0' for the correct operation of the device.

Table 40. FIFO_CTRL5 register description

ODR_FIFO_[3:0]	FIFO ODR selection, setting FIFO_MODE also. Default: 0000 For the configuration setting, refer to Table 41 .
FIFO_MODE_[2:0]	FIFO mode selection bits, setting ODR_FIFO also. Default value: 000 For the configuration setting, refer to Table 42 .

Table 41. FIFO ODR selection

ODR_FIFO_[3:0]	Configuration ⁽¹⁾
0000	FIFO disabled
0001	FIFO ODR is set to 12.5 Hz
0010	FIFO ODR is set to 26 Hz
0011	FIFO ODR is set to 52 Hz
0100	FIFO ODR is set to 104 Hz
0101	FIFO ODR is set to 208 Hz
0110	FIFO ODR is set to 416 Hz
0111	FIFO ODR is set to 833 Hz
1000	FIFO ODR is set to 1.66 kHz
1001	FIFO ODR is set to 3.33 kHz
1010	FIFO ODR is set to 6.66 kHz

1. If the device is working at an ODR slower than the one selected, FIFO ODR is limited to that ODR value. Moreover, these bits are effective if both the DATA_VALID_SEL_FIFO bit of [MASTER_CONFIG \(1Ah\)](#) and the TIMER_PEDO_FIFO_DRDY bit of [FIFO_CTRL2 \(07h\)](#) are set to 0.

Table 42. FIFO mode selection

FIFO_MODE_[2:0]	Configuration mode
000	Bypass mode. FIFO disabled.
001	FIFO mode. Stops collecting data when FIFO is full.
010	Reserved
011	Continuous mode until trigger is deasserted, then FIFO mode.
100	Bypass mode until trigger is deasserted, then Continuous mode.
101	Reserved
110	Continuous mode. If the FIFO is full, the new sample overwrites the older one.
111	Reserved

9.9 DRDY_PULSE_CFG_G (0Bh)

DataReady configuration register (r/w).

Table 43. DRDY_PULSE_CFG_G register

DRDY_PULSED	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	0 ⁽¹⁾	INT2_WRIST_TILT
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1. This bit must be set to '0' for the correct operation of the device.

Table 44. DRDY_PULSE_CFG_G register description

DRDY_PULSED	Enable pulsed DataReady mode. Default value: 0 (0: DataReady latched mode. Returns to 0 only after output data have been read; 1: DataReady pulsed mode. The DataReady pulses are 75 µs long.)
INT2_WRIST_TILT	Wrist tilt interrupt on INT2 pad. Default value: 0 (0: disabled; 1: enabled)

9.10 INT1_CTRL (0Dh)

INT1 pad control register (r/w).

Each bit in this register enables a signal to be carried through INT1. The pad's output will supply the OR combination of the selected signals.

Table 45. INT1_CTRL register

INT1_STEP_DETECTOR	INT1_SIGN_MOT	INT1_FULL_FLAG	INT1_FIFO_OVR	INT1_FTH	INT1_BOOT	INT1_DRDY_G	INT1_DRDY_XL
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Table 46. INT1_CTRL register description

INT1_STEP_DETECTOR	Pedometer step recognition interrupt enable on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT1_SIGN_MOT	Significant motion interrupt enable on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT1_FULL_FLAG	FIFO full flag interrupt enable on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT1_FIFO_OVR	FIFO overrun interrupt on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT1_FTH	FIFO threshold interrupt on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT1_BOOT	Boot status available on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT1_DRDY_G	Gyroscope Data Ready on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT1_DRDY_XL	Accelerometer Data Ready on INT1 pad. Default value: 0 (0: disabled; 1: enabled)

9.11 INT2_CTRL (0Eh)

INT2 pad control register (r/w).

Each bit in this register enables a signal to be carried through INT2. The pad's output will supply the OR combination of the selected signals.

Table 47. INT2_CTRL register

INT2_STEP_DELTA	INT2_STEP_COUNT_OV	INT2_FULL_FLAG	INT2_FIFO_OVR	INT2_FTH	INT2_DRDY_TEMP	INT2_DRDY_G	INT2_DRDY_XL
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Table 48. INT2_CTRL register description

INT2_STEP_DELTA	Pedometer step recognition interrupt on delta time ⁽¹⁾ enable on INT2 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_STEP_COUNT_OV	Step counter overflow interrupt enable on INT2 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_FULL_FLAG	FIFO full flag interrupt enable on INT2 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_FIFO_OVR	FIFO overrun interrupt on INT2 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_FTH	FIFO threshold interrupt on INT2 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_DRDY_TEMP	Temperature Data Ready on INT2 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_DRDY_G	Gyroscope Data Ready on INT2 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_DRDY_XL	Accelerometer Data Ready on INT2 pad. Default value: 0 (0: disabled; 1: enabled)

1. Delta time value is defined in register [STEP_COUNT_DELTA \(15h\)](#).

9.12 WHO_AM_I (0Fh)

Who_AM_I register (r). This register is a read-only register. Its value is fixed at 6Ah.

Table 49. WHO_AM_I register

0	1	1	0	1	0	1	0
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9.13 CTRL1_XL (10h)

Linear acceleration sensor control register 1 (r/w).

Table 50. CTRL1_XL register

ODR_XL3	ODR_XL2	ODR_XL1	ODR_XL0	FS_XL1	FS_XL0	LPF1_BW_SEL	BW0_XL
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Table 51. CTRL1_XL register description

ODR_XL [3:0]	Output data rate and power mode selection. Default value: 0000 (see Table 52).
FS_XL [1:0]	Accelerometer full-scale selection. Default value: 00. (00: ± 2 g; 01: ± 16 g; 10: ± 4 g; 11: ± 8 g)
LPF1_BW_SEL	Accelerometer digital LPF (LPF1) bandwidth selection. For bandwidth selection refer to CTRL8_XL (17h) .
BW0_XL	Accelerometer analog chain bandwidth selection (only for accelerometer ODR ≥ 1.67 kHz). (0: BW @ 1.5 kHz; 1: BW @ 400 Hz)

Table 52. Accelerometer ODR register setting

ODR_XL3	ODR_XL2	ODR_XL1	ODR_XL0	ODR selection [Hz] when XL_HM_MODE = 1	ODR selection [Hz] when XL_HM_MODE = 0
0	0	0	0	Power-down	Power-down
1	0	1	1	1.6 Hz (low power only)	12.5 Hz (high performance)
0	0	0	1	12.5 Hz (low power)	12.5 Hz (high performance)
0	0	1	0	26 Hz (low power)	26 Hz (high performance)
0	0	1	1	52 Hz (low power)	52 Hz (high performance)
0	1	0	0	104 Hz (normal mode)	104 Hz (high performance)
0	1	0	1	208 Hz (normal mode)	208 Hz (high performance)
0	1	1	0	416 Hz (high performance)	416 Hz (high performance)
0	1	1	1	833 Hz (high performance)	833 Hz (high performance)
1	0	0	0	1.66 kHz (high performance)	1.66 kHz (high performance)
1	0	0	1	3.33 kHz (high performance)	3.33 kHz (high performance)
1	0	1	0	6.66 kHz (high performance)	6.66 kHz (high performance)
1	1	x	x	Not allowed	Not allowed

9.14 CTRL2_G (11h)

Angular rate sensor control register 2 (r/w).

Table 53. CTRL2_G register

ODR_G3	ODR_G2	ODR_G1	ODR_G0	FS_G1	FS_G0	FS_125	0 ⁽¹⁾
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1. This bit must be set to '0' for the correct operation of the device.

Table 54. CTRL2_G register description

ODR_G [3:0]	Gyroscope output data rate selection. Default value: 0000 (Refer to Table 55)
FS_G [1:0]	Gyroscope full-scale selection. Default value: 00 (00: 250 dps; 01: 500 dps; 10: 1000 dps; 11: 2000 dps)
FS_125	Gyroscope full-scale at 125 dps. Default value: 0 (0: disabled; 1: enabled)

Table 55. Gyroscope ODR configuration setting

ODR_G3	ODR_G2	ODR_G1	ODR_G0	ODR [Hz] when G_HM_MODE = 1	ODR [Hz] when G_HM_MODE = 0
0	0	0	0	Power down	Power down
0	0	0	1	12.5 Hz (low power)	12.5 Hz (high performance)
0	0	1	0	26 Hz (low power)	26 Hz (high performance)
0	0	1	1	52 Hz (low power)	52 Hz (high performance)
0	1	0	0	104 Hz (normal mode)	104 Hz (high performance)
0	1	0	1	208 Hz (normal mode)	208 Hz (high performance)
0	1	1	0	416 Hz (high performance)	416 Hz (high performance)
0	1	1	1	833 Hz (high performance)	833 Hz (high performance)
1	0	0	0	1.66 kHz (high performance)	1.66 kHz (high performance)
1	0	0	1	3.33 kHz (high performance)	3.33 kHz (high performance)
1	0	1	0	6.66 kHz (high performance)	6.66 kHz (high performance)
1	0	1	1	Not available	Not available

9.15 CTRL3_C (12h)

Control register 3 (r/w).

Table 56. CTRL3_C register

BOOT	BDU	H_LACTIVE	PP_OD	SIM	IF_INC	BLE	SW_RESET
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Table 57. CTRL3_C register description

BOOT	Reboot memory content. Default value: 0 (0: normal mode; 1: reboot memory content)
BDU	Block Data Update. Default value: 0 (0: continuous update; 1: output registers not updated until MSB and LSB have been read)
H_LACTIVE	Interrupt activation level. Default value: 0 (0: interrupt output pads active high; 1: interrupt output pads active low)
PP_OD	Push-pull/open-drain selection on INT1 and INT2 pads. Default value: 0 (0: push-pull mode; 1: open-drain mode)
SIM	SPI Serial Interface Mode selection. Default value: 0 (0: 4-wire interface; 1: 3-wire interface).
IF_INC	Register address automatically incremented during a multiple byte access with a serial interface (I ² C or SPI). Default value: 1 (0: disabled; 1: enabled)
BLE	Big/Little Endian Data selection. Default value 0 (0: data LSB @ lower address; 1: data MSB @ lower address)
SW_RESET	Software reset. Default value: 0 (0: normal mode; 1: reset device) This bit is automatically cleared.

9.16 CTRL4_C (13h)

Control register 4 (r/w).

Table 58. CTRL4_C register

DEN_XL_EN	SLEEP	INT2_on_INT1	DEN_DRDY_INT1	DRDY_MASK	I2C_disable	LPF1_SEL_G	0 ⁽¹⁾
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1. This bit must be set to '0' for the correct operation of the device.

Table 59. CTRL4_C register description

DEN_XL_EN	Extend DEN functionality to accelerometer sensor. Default value: 0 (0: disabled; 1: enabled)
SLEEP	Gyroscope sleep mode enable. Default value: 0 (0: disabled; 1: enabled)
DEN_DRDY_INT1	DEN DRDY signal on INT1 pad. Default value: 0 (0: disabled; 1: enabled)
INT2_on_INT1	All interrupt signals available on INT1 pad enable. Default value: 0 (0: interrupt signals divided between INT1 and INT2 pads; 1: all interrupt signals in logic or on INT1 pad)
DRDY_MASK	Configuration 1 data available enable bit. Default value: 0 (0: DA timer disabled; 1: DA timer enabled)
I2C_disable	Disable I ² C interface. Default value: 0 (0: both I ² C and SPI enabled; 1: I ² C disabled, SPI only enabled)
LPF1_SEL_G	Enable gyroscope digital LPF1. The bandwidth can be selected through FTYPE[1:0] in CTRL6_C (15h) . (0: disabled; 1: enabled)

9.17 CTRL5_C (14h)

Control register 5 (r/w).

Table 60. CTRL5_C register

ROUNDING2	ROUNDING1	ROUNDING0	DEN_LH	ST1_G	ST0_G	ST1_XL	ST0_XL
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Table 61. CTRL5_C register description

ROUNDING[2:0]	Circular burst-mode (rounding) read from the output registers. Default value: 000 (000: no rounding; Others: refer to Table 62)
DEN_LH	DEN active level configuration. Default value: 0 (0: active low; 1: active high)
ST_G [1:0]	Angular rate sensor self-test enable. Default value: 00 (00: Self-test disabled; Other: refer to Table 63)
ST_XL [1:0]	Linear acceleration sensor self-test enable. Default value: 00 (00: Self-test disabled; Other: refer to Table 64)

Table 62. Output registers rounding pattern

ROUNDING[2:0]	Rounding pattern
000	No rounding
001	Accelerometer only
010	Gyroscope only
011	Gyroscope + accelerometer
100	Registers from SENSORHUB1_REG (2Eh) to SENSORHUB6_REG (33h) only
101	Accelerometer + registers from SENSORHUB1_REG (2Eh) to SENSORHUB6_REG (33h)
110	Gyroscope + accelerometer + registers from SENSORHUB1_REG (2Eh) to SENSORHUB6_REG (33h) and registers from SENSORHUB7_REG (34h) to SENSORHUB12_REG (39h)
111	Gyroscope + accelerometer + registers from SENSORHUB1_REG (2Eh) to SENSORHUB6_REG (33h)

Table 63. Angular rate sensor self-test mode selection

ST1_G	ST0_G	Self-test mode
0	0	Normal mode
0	1	Positive sign self-test
1	0	Not allowed
1	1	Negative sign self-test

Table 64. Linear acceleration sensor self-test mode selection

ST1_XL	ST0_XL	Self-test mode
0	0	Normal mode
0	1	Positive sign self-test
1	0	Negative sign self-test
1	1	Not allowed

9.18 CTRL6_C (15h)

Angular rate sensor control register 6 (r/w).

Table 65. CTRL6_C register

TRIG_EN	LVL_EN	LVL2_EN	XL_HM_MODE	USR_OFF_W	0 ⁽¹⁾	FTYPE_1	FTYPE_0
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1. This bit must be set to '0' for the correct operation of the device.

Table 66. CTRL6_C register description

TRIG_EN	DEN data edge-sensitive trigger enable. Refer to Table 67 .
LVL_EN	DEN data level-sensitive trigger enable. Refer to Table 67 .
LVL2_EN	DEN level-sensitive latched enable. Refer to Table 67 .
XL_HM_MODE	High-performance operating mode disable for accelerometer. Default value: 0 (0: high-performance operating mode enabled; 1: high-performance operating mode disabled)
USR_OFF_W	Weight of XL user offset bits of registers X_OFS_USR (73h) , Y_OFS_USR (74h) , Z_OFS_USR (75h) 0 = 2^{-10} g/LSB 1 = 2^{-6} g/LSB
FTYPE[1:0]	Gyroscope's low-pass filter (LPF1) bandwidth selection Table 68 shows the selectable bandwidth values.

Table 67. Trigger mode selection

TRIG_EN, LVL1_EN, LVL2_EN	Trigger mode
100	Edge-sensitive trigger mode is selected
010	Level-sensitive trigger mode is selected
011	Level-sensitive latched mode is selected
110	Level-sensitive FIFO enable mode is selected

Table 68. Gyroscope LPF1 bandwidth selection

FTYPE[1:0]	Bandwidth			
	ODR = 800 Hz	ODR = 1.6 kHz	ODR = 3.3 kHz	ODR = 6.6 kHz
00	245 Hz	315 Hz	343 Hz	351 Hz
01	195 Hz	224 Hz	234 Hz	237 Hz
10	155 Hz	168 Hz	172 Hz	173 Hz
11	293 Hz	505 Hz	925 Hz	937 Hz

9.19 CTRL7_G (16h)

Angular rate sensor control register 7 (r/w).

Table 69. CTRL7_G register

G_HM_MODE	HP_EN_G	HPM1_G	HPM0_G	0 ⁽¹⁾	ROUNDING_STATUS	0 ⁽¹⁾	0 ⁽¹⁾
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1. This bit must be set to '0' for the correct operation of the device.

Table 70. CTRL7_G register description

G_HM_MODE	High-performance operating mode disable for gyroscope(1). Default: 0 (0: high-performance operating mode enabled; 1: high-performance operating mode disabled)
HP_EN_G	Gyroscope digital high-pass filter enable. The filter is enabled only if the gyro is in HP mode. Default value: 0 (0: HPF disabled; 1: HPF enabled)
HPM_G[1:0]	Gyroscope digital HP filter cutoff selection. Default: 00 (00 = 16 mHz 01 = 65 mHz 10 = 260 mHz 11 = 1.04 Hz)
ROUNDING_STATUS	Source register rounding function on WAKE_UP_SRC (1Bh), TAP_SRC (1Ch), D6D_SRC (1Dh), STATUS_REG (1Eh), and FUNC_SRC (53h). Default value: 0 (0: Rounding disabled; 1: Rounding enabled)

9.20 CTRL8_XL (17h)

Linear acceleration sensor control register 8 (r/w).

Table 71. CTRL8_XL register

LPF2_XL_EN	HPCF_XL1	HPCF_XL0	HP_REF_MODE	INPUT_COMPOSITE	HP_SLOPE_XL_EN	0 ⁽¹⁾	LOW_PASS_ON_6D
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1. This bit must be set to '0' for the correct operation of the device.

Table 72. CTRL8_XL register description

LPF2_XL_EN	Accelerometer low-pass filter LPF2 selection. Refer to Figure 8 .
HPCF_XL[1:0]	Accelerometer LPF2 and high-pass filter configuration and cutoff setting. Refer to Table 73 .
HP_REF_MODE	Enable HP filter reference mode. Default value: 0 (0: disabled; 1: enabled ⁽¹⁾)
INPUT_COMPOSITE	Composite filter input selection. Default: 0 (0: ODR/2 low pass filtered sent to composite filter (default) 1: ODR/4 low pass filtered sent to composite filter)
HP_SLOPE_XL_EN	Accelerometer slope filter / high-pass filter selection. Refer to Figure 8 .
LOW_PASS_ON_6D	LPF2 on 6D function selection. Refer to Figure 8 .

1. When enabled, the first output data has to be discarded.

Table 73. Accelerometer bandwidth selection

HP_SLOPE_XL_EN	LPF2_XL_EN	LPF1_BW_SEL	HPCF_XL[1:0]	INPUT_COMPOSITE	Bandwidth
0 (low-pass path) ⁽¹⁾	0	0	-	-	ODR/2
		1	-	-	ODR/4
	1	-	00	1 (low noise) 0 (low latency)	ODR/50
			01		ODR/100
			10		ODR/9
			11		ODR/400
1 (high-pass path) ⁽²⁾	-	-	0	ODR/4	
				ODR/100	
				ODR/9	
				ODR/400	

1. The bandwidth column is related to LPF1 if LPF2_XL_EN = 0 or to LPF2 if LPF2_XL_EN = 1.

2. The bandwidth column is related to the slope filter if HPCF_XL[1:0] = 00 or to the HP filter if HPCF_XL[1:0] = 01/10/11.

9.21 CTRL9_XL (18h)

Linear acceleration sensor control register 9 (r/w).

Table 74. CTRL9_XL register

DEN_X	DEN_Y	DEN_Z	DEN_XL_G	0 ⁽¹⁾	SOFT_EN	0 ⁽¹⁾	0 ⁽¹⁾
-------	-------	-------	----------	------------------	---------	------------------	------------------

1. This bit must be set to '0' for the correct operation of the device.

Table 75. CTRL9_XL register description

DEN_X	DEN value stored in LSB of X-axis. Default value: 1 (0: DEN not stored in X-axis LSB; 1: DEN stored in X-axis LSB)
DEN_Y	DEN value stored in LSB of Y-axis. Default value: 1 (0: DEN not stored in Y-axis LSB; 1: DEN stored in Y-axis LSB)
DEN_Z	DEN value stored in LSB of Z-axis. Default value: 1 (0: DEN not stored in Z-axis LSB; 1: DEN stored in Z-axis LSB)
DEN_XL_G	DEN stamping sensor selection. Default value: 0 (0: DEN pin info stamped in the gyroscope axis selected by bits [7:5]; 1: DEN pin info stamped in the accelerometer axis selected by bits [7:5])
SOFT_EN	Enable soft-iron correction algorithm for magnetometer ⁽¹⁾ . Default value: 0 (0: soft-iron correction algorithm disabled; 1: soft-iron correction algorithm enabled)

1. This bit is effective if the IRON_EN bit of [MASTER_CONFIG \(1Ah\)](#) and the FUNC_EN bit of [CTRL10_C \(19h\)](#) are set to 1.

9.22 CTRL10_C (19h)

Control register 10 (r/w).

Table 76. CTRL10_C register

WRIST_TILT_EN	0 ⁽¹⁾	TIMER_EN	PEDO_EN	TILT_EN	FUNC_EN	PEDO_RST_STEP	SIGN_MOTION_EN
---------------	------------------	----------	---------	---------	---------	---------------	----------------

1. This bit must be set to '0' for the correct operation of the device.

Table 77. CTRL10_C register description

WRIST_TILT_EN	Enable wrist tilt algorithm ⁽¹⁾⁽²⁾ . Default value: 0 (0: wrist tilt algorithm disabled; 1: wrist tilt algorithm enabled)
TIMER_EN	Enable timestamp count. The count is saved in TIMESTAMP0_REG (40h) , TIMESTAMP1_REG (41h) and TIMESTAMP2_REG (42h) . Default: 0 (0: timestamp count disabled; 1: timestamp count enabled)
PEDO_EN	Enable pedometer algorithm ⁽¹⁾ . Default value: 0 (0: pedometer algorithm disabled; 1: pedometer algorithm enabled)
TILT_EN	Enable tilt calculation ⁽¹⁾ .
FUNC_EN	Enable embedded functionalities (pedometer, tilt, significant motion detection, sensor hub and ironing). Default value: 0 (0: disable functionalities of embedded functions and accelerometer filters; 1: enable functionalities of embedded functions and accelerometer filters)
PEDO_RST_STEP	Reset pedometer step counter. Default value: 0 (0: disabled; 1: enabled)
SIGN_MOTION_EN	Enable significant motion detection function ⁽¹⁾ . Default value: 0 (0: disabled; 1: enabled)

1. This is effective if the FUNC_EN bit is set to '1'.
2. By default, the wrist tilt algorithm is applied to the positive X-axis.

9.23 MASTER_CONFIG (1Ah)

Master configuration register (r/w).

Table 78. MASTER_CONFIG register

DRDY_ON_INT1	DATA_VALID_SEL_FIFO	0 ⁽¹⁾	START_CONFIG	PULL_UP_EN	PASS_THROUGH_MODE	IRON_EN	MASTER_ON
--------------	---------------------	------------------	--------------	------------	-------------------	---------	-----------

1. This bit must be set to '0' for the correct operation of the device.

Table 79. MASTER_CONFIG register description

DRDY_ON_INT1	Manage the Master DRDY signal on INT1 pad. Default: 0 (0: disable Master DRDY on INT1; 1: enable Master DRDY on INT1)
DATA_VALID_SEL_FIFO	Selection of FIFO data-valid signal. Default value: 0 (0: data-valid signal used to write data in FIFO is the XL/Gyro data-ready or step detection ⁽¹⁾ ; 1: data-valid signal used to write data in FIFO is the sensor hub data-ready)
START_CONFIG	Sensor Hub trigger signal selection. Default value: 0 (0: Sensor hub signal is the XL/Gyro data-ready; 1: Sensor hub signal external from INT2 pad.)
PULL_UP_EN	Auxiliary I ² C pull-up. Default value: 0 (0: internal pull-up on auxiliary I ² C line disabled; 1: internal pull-up on auxiliary I ² C line enabled)
PASS_THROUGH_MODE	I ² C interface pass-through. Default value: 0 (0: pass-through disabled; 1: pass-through enabled)
IRON_EN	Enable hard-iron correction algorithm for magnetometer ⁽²⁾ . Default value: 0 (0: hard-iron correction algorithm disabled; 1: hard-iron correction algorithm enabled)
MASTER_ON	Sensor hub I ² C master enable ⁽²⁾ . Default: 0 (0: master I ² C of sensor hub disabled; 1: master I ² C of sensor hub enabled)

1. If the TIMER_PEDO_FIFO_DRDY bit in [FIFO_CTRL2 \(07h\)](#) is set to 0, the trigger for writing data in FIFO is XL/Gyro data-ready, otherwise it's the step detection.

2. This is effective if the FUNC_EN bit is set to '1'.

9.24 WAKE_UP_SRC (1Bh)

Wake up interrupt source register (r).

Table 80. WAKE_UP_SRC register

0	0	FF_IA	SLEEP_STATE_IA	WU_IA	X_WU	Y_WU	Z_WU
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Table 81. WAKE_UP_SRC register description

FF_IA	Free-fall event detection status. Default: 0 (0: free-fall event not detected; 1: free-fall event detected)
SLEEP_STATE_IA	Sleep event status. Default value: 0 (0: sleep event not detected; 1: sleep event detected)
WU_IA	Wakeup event detection status. Default value: 0 (0: wakeup event not detected; 1: wakeup event detected.)
X_WU	Wakeup event detection status on X-axis. Default value: 0 (0: wakeup event on X-axis not detected; 1: wakeup event on X-axis detected)
Y_WU	Wakeup event detection status on Y-axis. Default value: 0 (0: wakeup event on Y-axis not detected; 1: wakeup event on Y-axis detected)
Z_WU	Wakeup event detection status on Z-axis. Default value: 0 (0: wakeup event on Z-axis not detected; 1: wakeup event on Z-axis detected)

9.25 TAP_SRC (1Ch)

Tap source register (r).

Table 82. TAP_SRC register

0	TAP_IA	SINGLE_TAP	DOUBLE_TAP	TAP_SIGN	X_TAP	Y_TAP	Z_TAP
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Table 83. TAP_SRC register description

TAP_IA	Tap event detection status. Default: 0 (0: tap event not detected; 1: tap event detected)
SINGLE_TAP	Single-tap event status. Default value: 0 (0: single tap event not detected; 1: single tap event detected)
DOUBLE_TAP	Double-tap event detection status. Default value: 0 (0: double-tap event not detected; 1: double-tap event detected.)
TAP_SIGN	Sign of acceleration detected by tap event. Default: 0 (0: positive sign of acceleration detected by tap event; 1: negative sign of acceleration detected by tap event)
X_TAP	Tap event detection status on X-axis. Default value: 0 (0: tap event on X-axis not detected; 1: tap event on X-axis detected)
Y_TAP	Tap event detection status on Y-axis. Default value: 0 (0: tap event on Y-axis not detected; 1: tap event on Y-axis detected)
Z_TAP	Tap event detection status on Z-axis. Default value: 0 (0: tap event on Z-axis not detected; 1: tap event on Z-axis detected)

9.26 D6D_SRC (1Dh)

Portrait, landscape, face-up and face-down source register (r)

Table 84. D6D_SRC register

DEN_ DRDY	D6D_IA	ZH	ZL	YH	YL	XH	XL
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Table 85. D6D_SRC register description

DEN_ DRDY	DEN data-ready signal. It is set high when data output is related to the data coming from a DEN active condition. ⁽¹⁾
D6D_ IA	Interrupt active for change position portrait, landscape, face-up, face-down. Default value: 0 (0: change position not detected; 1: change position detected)
ZH	Z-axis high event (over threshold). Default value: 0 (0: event not detected; 1: event (over threshold) detected)
ZL	Z-axis low event (under threshold). Default value: 0 (0: event not detected; 1: event (under threshold) detected)
YH	Y-axis high event (over threshold). Default value: 0 (0: event not detected; 1: event (over-threshold) detected)
YL	Y-axis low event (under threshold). Default value: 0 (0: event not detected; 1: event (under threshold) detected)
XH	X-axis high event (over threshold). Default value: 0 (0: event not detected; 1: event (over threshold) detected)
XL	X-axis low event (under threshold). Default value: 0 (0: event not detected; 1: event (under threshold) detected)

1. The DEN data-ready signal can be latched or pulsed depending on the value of the dataready_pulsed bit of the [DRDY_PULSE_CFG_G \(0Bh\)](#) register.

9.27 STATUS_REG (1Eh)

The STATUS_REG register is read by the SPI/I²C interface (r).

Table 86. STATUS_REG register

0	0	0	0	0	TDA	GDA	XLDA
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Table 87. STATUS_REG register description

TDA	Temperature new data available. Default: 0 (0: no set of data is available at temperature sensor output; 1: a new set of data is available at temperature sensor output)
GDA	Gyroscope new data available. Default value: 0 (0: no set of data available at gyroscope output; 1: a new set of data is available at gyroscope output)
XLDA	Accelerometer new data available. Default value: 0 (0: no set of data available at accelerometer output; 1: a new set of data is available at accelerometer output)

9.28 OUT_TEMP_L (20h), OUT_TEMP_H (21h)

Temperature data output register (r). L and H registers together express a 16-bit word in two's complement.

Table 88. OUT_TEMP_L register

Temp7	Temp6	Temp5	Temp4	Temp3	Temp2	Temp1	Temp0
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Table 89. OUT_TEMP_H register

Temp15	Temp14	Temp13	Temp12	Temp11	Temp10	Temp9	Temp8
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Table 90. OUT_TEMP register description

Temp[15:0]	Temperature sensor output data The value is expressed as two's complement sign extended on the MSB.
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9.29 OUTX_L_G (22h)

Angular rate sensor pitch axis (X) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2_G \(11h\)](#)) of the gyro user interface.

Table 91. OUTX_L_G register

D7	D6	D5	D4	D3	D2	D1	D0
----	----	----	----	----	----	----	----

Table 92. OUTX_L_G register description

D[7:0]	Pitch axis (X) angular rate value (LSbyte) D[15:0] expressed in two's complement and its value depends on the interface used: SPI1/I ² C: Gyro UI chain pitch axis output
--------	--

9.30 OUTX_H_G (23h)

Angular rate sensor pitch axis (X) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2_G \(11h\)](#)) of the gyro user interface.

Table 93. OUTX_H_G register

D15	D14	D13	D12	D11	D10	D9	D8
-----	-----	-----	-----	-----	-----	----	----

Table 94. OUTX_H_G register description

D[15:8]	Pitch axis (X) angular rate value (MSbyte) D[15:0] expressed in two's complement and its value depends on the interface used: SPI1/I ² C: Gyro UI chain pitch axis output
---------	--

9.31 OUTY_L_G (24h)

Angular rate sensor roll axis (Y) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2_G \(11h\)](#)) of the gyro user interface.

Table 95. OUTY_L_G register

D7	D6	D5	D4	D3	D2	D1	D0
----	----	----	----	----	----	----	----

Table 96. OUTY_L_G register description

D[7:0]	Roll axis (Y) angular rate value (LSbyte) D[15:0] expressed in two's complement and its value depends on the interface used: SPI1/I ² C: Gyro UI chain roll axis output
--------	--

9.32 OUTY_H_G (25h)

Angular rate sensor roll axis (Y) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2_G \(11h\)](#)) of the gyro user interface.

Table 97. OUTY_H_G register

D15	D14	D13	D12	D11	D10	D9	D8
-----	-----	-----	-----	-----	-----	----	----

Table 98. OUTY_H_G register description

D[15:8]	Roll axis (Y) angular rate value (MSbyte) D[15:0] expressed in two's complement and its value depends on the interface used: SPI1/I ² C: Gyro UI chain roll axis output
---------	--

9.33 OUTZ_L_G (26h)

Angular rate sensor yaw axis (Z) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2_G \(11h\)](#)) of the gyro user interface.

Table 99. OUTZ_L_G register

D7	D6	D5	D4	D3	D2	D1	D0
----	----	----	----	----	----	----	----

Table 100. OUTZ_L_G register description

D[7:0]	Yaw axis (Z) angular rate value (LSbyte) D[15:0] expressed in two's complement and its value depends on the interface used: SPI1/I ² C: Gyro UI chain yaw axis output
--------	--

9.34 OUTZ_H_G (27h)

Angular rate sensor Yaw axis (Z) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2_G \(11h\)](#)) of the gyro user interface.

Table 101. OUTZ_H_G register

D15	D14	D13	D12	D11	D10	D9	D8
-----	-----	-----	-----	-----	-----	----	----

Table 102. OUTZ_H_G register description

D[15:8]	Yaw axis (Z) angular rate value (MSbyte) D[15:0] expressed in two's complement and its value depends on the interface used: SPI1/I ² C: Gyro UI chain yaw axis output
---------	--

9.35 OUTX_L_XL (28h)

Linear acceleration sensor X-axis output register (r). The value is expressed as a 16-bit word in two's complement.

Table 103. OUTX_L_XL register

D7	D6	D5	D4	D3	D2	D1	D0
----	----	----	----	----	----	----	----

Table 104. OUTX_L_XL register description

D[7:0]	X-axis linear acceleration value (LSbyte)
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9.36 OUTX_H_XL (29h)

Linear acceleration sensor X-axis output register (r). The value is expressed as a 16-bit word in two's complement.

Table 105. OUTX_H_XL register

D15	D14	D13	D12	D11	D10	D9	D8
-----	-----	-----	-----	-----	-----	----	----

Table 106. OUTX_H_XL register description

D[15:8]	X-axis linear acceleration value (MSbyte)
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9.37 OUTY_L_XL (2Ah)

Linear acceleration sensor Y-axis output register (r). The value is expressed as a 16-bit word in two's complement.

Table 107. OUTY_L_XL register

D7	D6	D5	D4	D3	D2	D1	D0
----	----	----	----	----	----	----	----

Table 108. OUTY_L_XL register description

D[7:0]	Y-axis linear acceleration value (LSbyte)
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9.38 OUTY_H_XL (2Bh)

Linear acceleration sensor Y-axis output register (r). The value is expressed as a 16-bit word in two's complement.

Table 109. OUTY_H_G register

D15	D14	D13	D12	D11	D10	D9	D8
-----	-----	-----	-----	-----	-----	----	----

Table 110. OUTY_H_G register description

D[15:8]	Y-axis linear acceleration value (MSbyte)
---------	---

9.39 OUTZ_L_XL (2Ch)

Linear acceleration sensor Z-axis output register (r). The value is expressed as a 16-bit word in two's complement.

Table 111. OUTZ_L_XL register

D7	D6	D5	D4	D3	D2	D1	D0
----	----	----	----	----	----	----	----

Table 112. OUTZ_L_XL register description

D[7:0]	Z-axis linear acceleration value (LSbyte)
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9.40 OUTZ_H_XL (2Dh)

Linear acceleration sensor Z-axis output register (r). The value is expressed as a 16-bit word in two's complement.

Table 113. OUTZ_H_XL register

D15	D14	D13	D12	D11	D10	D9	D8
-----	-----	-----	-----	-----	-----	----	----

Table 114. OUTZ_H_XL register description

D[15:8]	Z-axis linear acceleration value (MSbyte)
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9.41 SENSORHUB1_REG (2Eh)

First byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 115. SENSORHUB1_REG register

SHub1_7	SHub1_6	SHub1_5	SHub1_4	SHub1_3	SHub1_2	SHub1_1	SHub1_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 116. SENSORHUB1_REG register description

SHub1_[7:0]	First byte associated to external sensors
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9.42 SENSORHUB2_REG (2Fh)

Second byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operations configurations (for external sensors from x = 0 to x = 3).

Table 117. SENSORHUB2_REG register

SHub2_7	SHub2_6	SHub2_5	SHub2_4	SHub2_3	SHub2_2	SHub2_1	SHub2_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 118. SENSORHUB2_REG register description

SHub2_[7:0]	Second byte associated to external sensors
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9.43 SENSORHUB3_REG (30h)

Third byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operations configurations (for external sensors from x = 0 to x = 3).

Table 119. SENSORHUB3_REG register

SHub3_7	SHub3_6	SHub3_5	SHub3_4	SHub3_3	SHub3_2	SHub3_1	SHub3_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 120. SENSORHUB3_REG register description

SHub3_[7:0]	Third byte associated to external sensors
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9.44 SENSORHUB4_REG (31h)

Fourth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 121. SENSORHUB4_REG register

SHub4_7	SHub4_6	SHub4_5	SHub4_4	SHub4_3	SHub4_2	SHub4_1	SHub4_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 122. SENSORHUB4_REG register description

SHub4_[7:0]	Fourth byte associated to external sensors
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9.45 SENSORHUB5_REG (32h)

Fifth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 123. SENSORHUB5_REG register

SHub5_7	SHub5_6	SHub5_5	SHub5_4	SHub5_3	SHub5_2	SHub5_1	SHub5_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 124. SENSORHUB5_REG register description

SHub5_[7:0]	Fifth byte associated to external sensors
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9.46 SENSORHUB6_REG (33h)

Sixth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 125. SENSORHUB6_REG register

SHub6_7	SHub6_6	SHub6_5	SHub6_4	SHub6_3	SHub6_2	SHub6_1	SHub6_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 126. SENSORHUB6_REG register description

SHub6_[7:0]	Sixth byte associated to external sensors
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9.47 SENSORHUB7_REG (34h)

Seventh byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 127. SENSORHUB7_REG register

SHub7_7	SHub7_6	SHub7_5	SHub7_4	SHub7_3	SHub7_2	SHub7_1	SHub7_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 128. SENSORHUB7_REG register description

SHub7_[7:0]	Seventh byte associated to external sensors
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9.48 SENSORHUB8_REG(35h)

Eighth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 129. SENSORHUB8_REG register

SHub8_7	SHub8_6	SHub8_5	SHub8_4	SHub8_3	SHub8_2	SHub8_1	SHub8_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 130. SENSORHUB8_REG register description

SHub8_[7:0]	Eighth byte associated to external sensors
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9.49 SENSORHUB9_REG (36h)

Ninth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 131. SENSORHUB9_REG register

SHub9_7	SHub9_6	SHub9_5	SHub9_4	SHub9_3	SHub9_2	SHub9_1	SHub9_0
---------	---------	---------	---------	---------	---------	---------	---------

Table 132. SENSORHUB9_REG register description

SHub9_[7:0]	Ninth byte associated to external sensors
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9.50 SENSORHUB10_REG (37h)

Tenth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 133. SENSORHUB10_REG register

SHub10_7	SHub10_6	SHub10_5	SHub10_4	SHub10_3	SHub10_2	SHub10_1	SHub10_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 134. SENSORHUB10_REG register description

SHub10_[7:0]	Tenth byte associated to external sensors
--------------	---

9.51 SENSORHUB11_REG (38h)

Eleventh byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 135. SENSORHUB11_REG register

SHub11_7	SHub11_6	SHub11_5	SHub11_4	SHub11_3	SHub11_2	SHub11_1	SHub11_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 136. SENSORHUB11_REG register description

SHub11_[7:0]	Eleventh byte associated to external sensors
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9.52 SENSORHUB12_REG (39h)

Twelfth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 137. SENSORHUB12_REG register

SHub12_7	SHub12_6	SHub12_5	SHub12_4	SHub12_3	SHub12_2	SHub12_1	SHub12_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 138. SENSORHUB12_REG register description

SHub12[7:0]	Twelfth byte associated to external sensors
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9.53 FIFO_STATUS1 (3Ah)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3_C \(12h\)](#) to 1.

Table 139. FIFO_STATUS1 register

DIFF_FIFO_7	DIFF_FIFO_6	DIFF_FIFO_5	DIFF_FIFO_4	DIFF_FIFO_3	DIFF_FIFO_2	DIFF_FIFO_1	DIFF_FIFO_0
-------------	-------------	-------------	-------------	-------------	-------------	-------------	-------------

Table 140. FIFO_STATUS1 register description

DIFF_FIFO_[7:0]	Number of unread words (16-bit axes) stored in FIFO ⁽¹⁾ .
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1. For a complete number of unread samples, consider DIFF_FIFO [10:8] in [FIFO_STATUS2 \(3Bh\)](#).

9.54 FIFO_STATUS2 (3Bh)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3_C \(12h\)](#) to 1.

Table 141. FIFO_STATUS2 register

WaterM	OVER_RUN	FIFO_ FULL_ SMART	FIFO_ EMPTY	0	DIFF_ FIFO_10	DIFF_ FIFO_9	DIFF_ FIFO_8
--------	----------	-------------------	-------------	---	---------------	--------------	--------------

Table 142. FIFO_STATUS2 register description

WaterM	FIFO watermark status. The watermark is set through bits FTH_[7:0] in FIFO_CTRL1 (06h) . Default value: 0 (0: FIFO filling is lower than watermark level ⁽¹⁾ ; 1: FIFO filling is equal to or higher than the watermark level)
OVER_RUN	FIFO overrun status. Default value: 0 (0: FIFO is not completely filled; 1: FIFO is completely filled)
FIFO_FULL_ SMART	Smart FIFO full status. Default value: 0 (0: FIFO is not full; 1: FIFO will be full at the next ODR)
FIFO_EMPTY	FIFO empty bit. Default value: 0 (0: FIFO contains data; 1: FIFO is empty)
DIFF_FIFO_[10:8]	Number of unread words (16-bit axes) stored in FIFO ⁽²⁾ .

1. FIFO watermark level is set in FTH_[10:0] in [FIFO_CTRL1 \(06h\)](#) and [FIFO_CTRL2 \(07h\)](#).

2. For a complete number of unread samples, consider DIFF_FIFO [7:0] in [FIFO_STATUS1 \(3Ah\)](#).

9.55 FIFO_STATUS3 (3Ch)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3_C \(12h\)](#) to 1.

Table 143. FIFO_STATUS3 register

FIFO_ PATTERN _7	FIFO_ PATTERN _6	FIFO_ PATTERN _5	FIFO_ PATTERN _4	FIFO_ PATTERN _3	FIFO_ PATTERN _2	FIFO_ PATTERN _1	FIFO_ PATTERN _0
------------------	------------------	------------------	------------------	------------------	------------------	------------------	------------------

Table 144. FIFO_STATUS3 register description

FIFO_ PATTERN_[7:0]	Word of recursive pattern read at the next reading.
---------------------	---

9.56 FIFO_STATUS4 (3Dh)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3_C \(12h\)](#) to 1.

Table 145. FIFO_STATUS4 register

0	0	0	0	0	0	FIFO_PATTERN_9	FIFO_PATTERN_8
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Table 146. FIFO_STATUS4 register description

FIFO_PATTERN_[9:8]	Word of recursive pattern read at the next reading.
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9.57 FIFO_DATA_OUT_L (3Eh)

FIFO data output register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3_C \(12h\)](#) to 1.

Table 147. FIFO_DATA_OUT_L register

DATA_OUT_7 FIFO_L_7	DATA_OUT_6 FIFO_L_6	DATA_OUT_5 FIFO_L_5	DATA_OUT_4 FIFO_L_4	DATA_OUT_3 FIFO_L_3	DATA_OUT_2 FIFO_L_2	DATA_OUT_1 FIFO_L_1	DATA_OUT_0 FIFO_L_0
------------------------	------------------------	------------------------	------------------------	------------------------	------------------------	------------------------	------------------------

Table 148. FIFO_DATA_OUT_L register description

DATA_OUT_FIFO_L_[7:0]	FIFO data output (first byte)
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9.58 FIFO_DATA_OUT_H (3Fh)

FIFO data output register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3_C \(12h\)](#) to 1.

Table 149. FIFO_DATA_OUT_H register

DATA_OUT_7 FIFO_H_7	DATA_OUT_6 FIFO_H_6	DATA_OUT_5 FIFO_H_5	DATA_OUT_4 FIFO_H_4	DATA_OUT_3 FIFO_H_3	DATA_OUT_2 FIFO_H_2	DATA_OUT_1 FIFO_H_1	DATA_OUT_0 FIFO_H_0
------------------------	------------------------	------------------------	------------------------	------------------------	------------------------	------------------------	------------------------

Table 150. FIFO_DATA_OUT_H register description

DATA_OUT_FIFO_H_[7:0]	FIFO data output (second byte)
-----------------------	--------------------------------

9.59 TIMESTAMP0_REG (40h)

Timestamp first (least significant) byte data output register (r). The value is expressed as a 24-bit word and the bit resolution is defined by setting the value in [WAKE_UP_DUR \(5Ch\)](#).

Table 151. TIMESTAMP0_REG register

TIMESTA MP0_7	TIMESTA MP0_6	TIMESTA MP0_5	TIMESTA MP0_4	TIMESTA MP0_3	TIMESTA MP0_2	TIMESTA MP0_1	TIMESTA MP0_0
------------------	------------------	------------------	------------------	------------------	------------------	------------------	------------------

Table 152. TIMESTAMP0_REG register description

TIMESTAMP0_[7:0]	TIMESTAMP first byte data output
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9.60 TIMESTAMP1_REG (41h)

Timestamp second byte data output register (r). The value is expressed as a 24-bit word and the bit resolution is defined by setting value in [WAKE_UP_DUR \(5Ch\)](#).

Table 153. TIMESTAMP1_REG register

TIMESTA MP1_7	TIMESTA MP1_6	TIMESTA MP1_5	TIMESTA MP1_4	TIMESTA MP1_3	TIMESTA MP1_2	TIMESTA MP1_1	TIMESTA MP1_0
------------------	------------------	------------------	------------------	------------------	------------------	------------------	------------------

Table 154. TIMESTAMP1_REG register description

TIMESTAMP1_[7:0]	TIMESTAMP second byte data output
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9.61 TIMESTAMP2_REG (42h)

Timestamp third (most significant) byte data output register (r/w). The value is expressed as a 24-bit word and the bit resolution is defined by setting the value in [WAKE_UP_DUR \(5Ch\)](#). To reset the timer, the AAh value has to be stored in this register.

Table 155. TIMESTAMP2_REG register

TIMESTA MP2_7	TIMESTA MP2_6	TIMESTA MP2_5	TIMESTA MP2_4	TIMESTA MP2_3	TIMESTA MP2_2	TIMESTA MP2_1	TIMESTA MP2_0
------------------	------------------	------------------	------------------	------------------	------------------	------------------	------------------

Table 156. TIMESTAMP2_REG register description

TIMESTAMP2_[7:0]	TIMESTAMP third byte data output
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9.62 STEP_TIMESTAMP_L (49h)

Step counter timestamp information register (r). When a step is detected, the value of TIMESTAMP_REG1 register is copied in STEP_TIMESTAMP_L.

Table 157. STEP_TIMESTAMP_L register

STEP_TIMESTAMP_L_7	STEP_TIMESTAMP_L_6	STEP_TIMESTAMP_L_5	STEP_TIMESTAMP_L_4	STEP_TIMESTAMP_L_3	STEP_TIMESTAMP_L_2	STEP_TIMESTAMP_L_1	STEP_TIMESTAMP_L_0
--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------

Table 158. STEP_TIMESTAMP_L register description

STEP_TIMESTAMP_L[7:0]	Timestamp of last step detected.
-----------------------	----------------------------------

9.63 STEP_TIMESTAMP_H (4Ah)

Step counter timestamp information register (r). When a step is detected, the value of TIMESTAMP_REG2 register is copied in STEP_TIMESTAMP_H.

Table 159. STEP_TIMESTAMP_H register

STEP_TIMESTAMP_H_7	STEP_TIMESTAMP_H_6	STEP_TIMESTAMP_H_5	STEP_TIMESTAMP_H_4	STEP_TIMESTAMP_H_3	STEP_TIMESTAMP_H_2	STEP_TIMESTAMP_H_1	STEP_TIMESTAMP_H_0
--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------

Table 160. STEP_TIMESTAMP_H register description

STEP_TIMESTAMP_H[7:0]	Timestamp of last step detected.
-----------------------	----------------------------------

9.64 STEP_COUNTER_L (4Bh)

Step counter output register (r).

Table 161. STEP_COUNTER_L register

STEP_COUNTER_L_7	STEP_COUNTER_L_6	STEP_COUNTER_L_5	STEP_COUNTER_L_4	STEP_COUNTER_L_3	STEP_COUNTER_L_2	STEP_COUNTER_L_1	STEP_COUNTER_L_0
------------------	------------------	------------------	------------------	------------------	------------------	------------------	------------------

Table 162. STEP_COUNTER_L register description

STEP_COUNTER_L[7:0]	Step counter output (LSbyte)
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9.65 STEP_COUNTER_H (4Ch)

Step counter output register (r).

Table 163. STEP_COUNTER_H register

STEP_CO UNTER_H _7	STEP_CO UNTER_H _6	STEP_CO UNTER_H _5	STEP_CO UNTER_H _4	STEP_CO UNTER_H _3	STEP_CO UNTER_H _2	STEP_CO UNTER_H _1	STEP_CO UNTER_H _0
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Table 164. STEP_COUNTER_H register description

STEP_COUNTER_H[7:0]	Step counter output (MSbyte)
---------------------	------------------------------

9.66 SENSORHUB13_REG (4Dh)

Thirteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 165. SENSORHUB13_REG register

SHub13_7	SHub13_6	SHub13_5	SHub13_4	SHub13_3	SHub13_2	SHub13_1	SHub13_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 166. SENSORHUB13_REG register description

SHub13[7:0]	Thirteenth byte associated to external sensors
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9.67 SENSORHUB14_REG (4Eh)

Fourteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 167. SENSORHUB14_REG register

SHub14_7	SHub14_6	SHub14_5	SHub14_4	SHub14_3	SHub14_2	SHub14_1	SHub14_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 168. SENSORHUB14_REG register description

SHub14[7:0]	Fourteenth byte associated to external sensors
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9.68 SENSORHUB15_REG (4Fh)

Fifteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 169. SENSORHUB15_REG register

SHub15_7	SHub15_6	SHub15_5	SHub15_4	SHub15_3	SHub15_2	SHub15_1	SHub15_0
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Table 170. SENSORHUB15_REG register description

SHub15[7:0]	Fifteenth byte associated to external sensors
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9.69 SENSORHUB16_REG (50h)

Sixteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 171. SENSORHUB16_REG register

SHub16_7	SHub16_6	SHub16_5	SHub16_4	SHub16_3	SHub16_2	SHub16_1	SHub16_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 172. SENSORHUB16_REG register description

SHub16_[7:0]	Sixteenth byte associated to external sensors
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9.70 SENSORHUB17_REG (51h)

Seventeenth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 173. SENSORHUB17_REG register

SHub17_7	SHub17_6	SHub17_5	SHub17_4	SHub17_3	SHub17_2	SHub17_1	SHub17_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 174. SENSORHUB17_REG register description

SHub17_[7:0]	Seventeenth byte associated to external sensors
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9.71 SENSORHUB18_REG (52h)

Eighteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

Table 175. SENSORHUB18_REG register

SHub18_7	SHub18_6	SHub18_5	SHub18_4	SHub18_3	SHub18_2	SHub18_1	SHub18_0
----------	----------	----------	----------	----------	----------	----------	----------

Table 176. SENSORHUB18_REG register description

SHub18_[7:0]	Eighteenth byte associated to external sensors
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9.72 FUNC_SRC1 (53h)

Significant motion, tilt, step detector, hard/soft-iron and sensor hub interrupt source register (r).

Table 177. FUNC_SRC1 register

STEP_COUNT_DELTA_IA	SIGN_MOTION_IA	TILT_IA	STEP_DETECTED	STEP_OVERFLOW	HI_FAIL	SI_END_OP	SENSOR_HUB_END_OP
---------------------	----------------	---------	---------------	---------------	---------	-----------	-------------------

Table 178. FUNC_SRC1 register description

STEP_COUNT_DELTA_IA	Pedometer step recognition on delta time status. Default value: 0 (0: no step recognized during delta time; 1: at least one step recognized during delta time)
SIGN_MOTION_IA	Significant motion event detection status. Default value: 0 (0: significant motion event not detected; 1: significant motion event detected)
TILT_IA	Tilt event detection status. Default value: 0 (0: tilt event not detected; 1: tilt event detected)
STEP_DETECTED	Step detector event detection status. Default value: 0 (0: step detector event not detected; 1: step detector event detected)
STEP_OVERFLOW	Step counter overflow status. Default value: 0 (0: step counter value < 2 ¹⁶ ; 1: step counter value reached 2 ¹⁶)
HI_FAIL	Fail in hard/soft-ironing algorithm.
SI_END_OP	Hard/soft-iron calculation status. Default value: 0 (0: Hard/soft-iron calculation not concluded; 1: Hard/soft-iron calculation concluded)
SENSORHUB_END_OP	Sensor hub communication status. Default value: 0 (0: sensor hub communication not concluded; 1: sensor hub communication concluded)

9.73 FUNC_SRC2 (54h)

Wrist tilt interrupt source register (r).

Table 179. FUNC_SRC2 register

0	SLAVE3_NACK	SLAVE2_NACK	SLAVE1_NACK	SLAVE0_NACK	0	0	WRIST_TILT_IA
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Table 180. FUNC_SRC2 register description

SLAVE3_NACK	This bit is set to 1 if Not acknowledge occurs on slave 3 communication. Default value: 0
SLAVE2_NACK	This bit is set to 1 if Not acknowledge occurs on slave 2 communication. Default value: 0
SLAVE1_NACK	This bit is set to 1 if Not acknowledge occurs on slave 1 communication. Default value: 0
SLAVE0_NACK	This bit is set to 1 if Not acknowledge occurs on slave 0 communication. Default value: 0
WRIST_TILT_IA	Wrist tilt event detection status. Default value: 0 (0: Wrist tilt event not detected; 1: Wrist tilt event detected)

9.74 WRIST_TILT_IA (55h)

Wrist tilt interrupt source register (r).

Table 181. WRIST_TILT_IA register

WRIST_ TILT_IA_ Xpos	WRIST_ TILT_IA_ Xneg	WRIST_ TILT_IA_ Ypos	WRIST_ TILT_IA_ Yneg	WRIST_ TILT_IA_ Zpos	WRIST_ TILT_IA_ Zneg	0	0
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Table 182. WRIST_TILT_IA register description

WRIST_ TILT_IA_ Xpos	Absolute Wrist Tilt event detection status on X-positive axis. Default value: 0 (0: Absolute Wrist Tilt event on X-positive axis not detected; 1: Absolute Wrist Tilt event on X-positive axis detected)
WRIST_ TILT_IA_ Xneg	Absolute Wrist Tilt event detection status on X-negative axis. Default value: 0 (0: Absolute Wrist Tilt event on X-negative axis not detected; 1: Absolute Wrist Tilt event on X-negative axis detected)
WRIST_ TILT_IA_ Ypos	Absolute Wrist Tilt event detection status on Y-positive axis. Default value: 0 (0: Absolute Wrist Tilt event on Y-positive axis not detected; 1: Absolute Wrist Tilt event on Y-positive axis detected)
WRIST_ TILT_IA_ Yneg	Absolute Wrist Tilt event detection status on Y-negative axis. Default value: 0 (0: Absolute Wrist Tilt event on Y-negative axis not detected; 1: Absolute Wrist Tilt event on Y-negative axis detected)
WRIST_ TILT_IA_ Zpos	Absolute Wrist Tilt event detection status on Z-positive axis. Default value: 0 (0: Absolute Wrist Tilt event on Z-positive axis not detected; 1: Absolute Wrist Tilt event on Z-positive axis detected)
WRIST_ TILT_IA_ Zneg	Absolute Wrist Tilt event detection status on Z-negative axis. Default value: 0 (0: Absolute Wrist Tilt event on Z-negative axis not detected; 1: Absolute Wrist Tilt event on Z-negative axis detected)

9.75 TAP_CFG (58h)

Enables interrupt and inactivity functions, configuration of filtering and tap recognition functions (r/w).

Table 183. TAP_CFG register

INTERRUPTS_ENABLE	INACT_EN1	INACT_EN0	SLOPE_FDS	TAP_X_EN	TAP_Y_EN	TAP_Z_EN	LIR
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Table 184. TAP_CFG register description

INTERRUPTS_ENABLE	Enable basic interrupts (6D/4D, free-fall, wake-up, tap, inactivity). Default 0. (0: interrupt disabled; 1: interrupt enabled)
INACT_EN[1:0]	Enable inactivity function. Default value: 00 (00: disabled 01: sets accelerometer ODR to 12.5 Hz (low-power mode), gyro does not change; 10: sets accelerometer ODR to 12.5 Hz (low-power mode), gyro to sleep mode; 11: sets accelerometer ODR to 12.5 Hz (low-power mode), gyro to power-down mode)
SLOPE_FDS	HPF or SLOPE filter selection on wake-up and Activity/Inactivity functions. Refer to Figure 8 . Default value: 0 (0: SLOPE filter applied; 1: HPF applied)
TAP_X_EN	Enable X direction in tap recognition. Default value: 0 (0: X direction disabled; 1: X direction enabled)
TAP_Y_EN	Enable Y direction in tap recognition. Default value: 0 (0: Y direction disabled; 1: Y direction enabled)
TAP_Z_EN	Enable Z direction in tap recognition. Default value: 0 (0: Z direction disabled; 1: Z direction enabled)
LIR	Latched Interrupt. Default value: 0 (0: interrupt request not latched; 1: interrupt request latched)

9.76 TAP_THS_6D (59h)

Portrait/landscape position and tap function threshold register (r/w).

Table 185. TAP_THS_6D register

D4D_EN	SIXD_THS 1	SIXD_THS 0	TAP_THS 4	TAP_THS 3	TAP_THS 2	TAP_THS 1	TAP_THS 0
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Table 186. TAP_THS_6D register description

D4D_EN	4D orientation detection enable. Z-axis position detection is disabled. Default value: 0 (0: enabled; 1: disabled)
SIXD_THS[1:0]	Threshold for 4D/6D function. Default value: 00 For details, refer to Table 187 .
TAP_THS[4:0]	Threshold for tap recognition. Default value: 00000 1 LSB corresponds to $FS_{XL}/2^5$

Table 187. Threshold for D4D/D6D function

SIXD_THS[1:0]	Threshold value
00	80 degrees
01	70 degrees
10	60 degrees
11	50 degrees

9.77 INT_DUR2 (5Ah)

Tap recognition function setting register (r/w).

Table 188. INT_DUR2 register

DUR3	DUR2	DUR1	DUR0	QUIET1	QUIET0	SHOCK1	SHOCK0
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Table 189. INT_DUR2 register description

DUR[3:0]	Duration of maximum time gap for double tap recognition. Default: 0000 When double tap recognition is enabled, this register expresses the maximum time between two consecutive detected taps to determine a double tap event. The default value of these bits is 0000b which corresponds to $16 \times ODR_{XL}$ time. If the DUR[3:0] bits are set to a different value, 1LSB corresponds to $32 \times ODR_{XL}$ time.
QUIET[1:0]	Expected quiet time after a tap detection. Default value: 00 Quiet time is the time after the first detected tap in which there must not be any overthreshold event. The default value of these bits is 00b which corresponds to $2 \times ODR_{XL}$ time. If the QUIET[1:0] bits are set to a different value, 1LSB corresponds to $4 \times ODR_{XL}$ time.
SHOCK[1:0]	Maximum duration of overthreshold event. Default value: 00 Maximum duration is the maximum time of an overthreshold signal detection to be recognized as a tap event. The default value of these bits is 00b which corresponds to $4 \times ODR_{XL}$ time. If the SHOCK[1:0] bits are set to a different value, 1LSB corresponds to $8 \times ODR_{XL}$ time.

9.78 WAKE_UP_THS (5Bh)

Single and double-tap function threshold register (r/w).

Table 190. WAKE_UP_THS register

SINGLE_DOUBLE_TAP	0	WK_THS5	WK_THS4	WK_THS3	WK_THS2	WK_THS1	WK_THS0
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Table 191. WAKE_UP_THS register description

SINGLE_DOUBLE_TAP	Single/double-tap event enable. Default: 0 (0: only single-tap event enabled; 1: both single and double-tap events enabled)
WK_THS[5:0]	Threshold for wakeup. Default value: 000000 1 LSb corresponds to $FS_{XL}/2^6$

9.79 WAKE_UP_DUR (5Ch)

Free-fall, wakeup, timestamp and sleep mode functions duration setting register (r/w).

Table 192. WAKE_UP_DUR register

FF_DUR5	WAKE_DUR1	WAKE_DUR0	TIMER_HR	SLEEP_DUR3	SLEEP_DUR2	SLEEP_DUR1	SLEEP_DUR0
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Table 193. WAKE_UP_DUR register description

FF_DUR5	Free fall duration event. Default: 0 For the complete configuration of the free-fall duration, refer to FF_DUR[4:0] in FREE_FALL (5Dh) configuration. 1 LSB = 1 ODR_time
WAKE_DUR[1:0]	Wake up duration event. Default: 00 1LSB = 1 ODR_time
TIMER_HR	Timestamp register resolution setting ⁽¹⁾ . Default value: 0 (0: 1LSB = 6.4 ms; 1: 1LSB = 25 μ s)
SLEEP_DUR[3:0]	Duration to go in sleep mode. Default value: 0000 (this corresponds to 16 ODR) 1 LSB = 512 ODR

1. Configuration of this bit affects [TIMESTAMP0_REG \(40h\)](#), [TIMESTAMP1_REG \(41h\)](#), [TIMESTAMP2_REG \(42h\)](#), [STEP_TIMESTAMP_L \(49h\)](#), [STEP_TIMESTAMP_H \(4Ah\)](#), and [CTRL6_C \(15h\)](#) registers.

9.80 FREE_FALL (5Dh)

Free-fall function duration setting register (r/w).

Table 194. FREE_FALL register

FF_DUR4	FF_DUR3	FF_DUR2	FF_DUR1	FF_DUR0	FF_THS2	FF_THS1	FF_THS0
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Table 195. FREE_FALL register description

FF_DUR[4:0]	Free-fall duration event. Default: 0 For the complete configuration of the free fall duration, refer to FF_DUR5 in WAKE_UP_DUR (5Ch) configuration
FF_THS[2:0]	Free fall threshold setting. Default: 000 For details refer to Table 196 .

Table 196. Threshold for free-fall function

FF_THS[2:0]	Threshold value
000	156 mg
001	219 mg
010	250 mg
011	312 mg
100	344 mg
101	406 mg
110	469 mg
111	500 mg

9.81 MD1_CFG (5Eh)

Functions routing on INT1 register (r/w).

Table 197. MD1_CFG register

INT1_INACT_STATE	INT1_SINGLE_TAP	INT1_WU	INT1_FF	INT1_DOUBLE_TAP	INT1_6D	INT1_TILT	INT1_TIMER
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Table 198. MD1_CFG register description

INT1_INACT_STATE	Routing on INT1 of inactivity mode. Default: 0 (0: routing on INT1 of inactivity disabled; 1: routing on INT1 of inactivity enabled)
INT1_SINGLE_TAP	Single-tap recognition routing on INT1. Default: 0 (0: routing of single-tap event on INT1 disabled; 1: routing of single-tap event on INT1 enabled)
INT1_WU	Routing of wakeup event on INT1. Default value: 0 (0: routing of wakeup event on INT1 disabled; 1: routing of wakeup event on INT1 enabled)
INT1_FF	Routing of free-fall event on INT1. Default value: 0 (0: routing of free-fall event on INT1 disabled; 1: routing of free-fall event on INT1 enabled)
INT1_DOUBLE_TAP	Routing of tap event on INT1. Default value: 0 (0: routing of double-tap event on INT1 disabled; 1: routing of double-tap event on INT1 enabled)
INT1_6D	Routing of 6D event on INT1. Default value: 0 (0: routing of 6D event on INT1 disabled; 1: routing of 6D event on INT1 enabled)
INT1_TILT	Routing of tilt event on INT1. Default value: 0 (0: routing of tilt event on INT1 disabled; 1: routing of tilt event on INT1 enabled)
INT1_TIMER	Routing of end counter event of timer on INT1. Default value: 0 (0: routing of end counter event of timer on INT1 disabled; 1: routing of end counter event of timer event on INT1 enabled)

9.82 MD2_CFG (5Fh)

Functions routing on INT2 register (r/w).

Table 199. MD2_CFG register

INT2_INACT_STATE	INT2_SINGLE_TAP	INT2_WU	INT2_FF	INT2_DOUBLE_TAP	INT2_6D	INT2_TILT	INT2_IRON
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Table 200. MD2_CFG register description

INT2_INACT_STATE	Routing on INT2 of inactivity mode. Default: 0 (0: routing on INT2 of inactivity disabled; 1: routing on INT2 of inactivity enabled)
INT2_SINGLE_TAP	Single-tap recognition routing on INT2. Default: 0 (0: routing of single-tap event on INT2 disabled; 1: routing of single-tap event on INT2 enabled)
INT2_WU	Routing of wakeup event on INT2. Default value: 0 (0: routing of wakeup event on INT2 disabled; 1: routing of wake-up event on INT2 enabled)
INT2_FF	Routing of free-fall event on INT2. Default value: 0 (0: routing of free-fall event on INT2 disabled; 1: routing of free-fall event on INT2 enabled)
INT2_DOUBLE_TAP	Routing of tap event on INT2. Default value: 0 (0: routing of double-tap event on INT2 disabled; 1: routing of double-tap event on INT2 enabled)
INT2_6D	Routing of 6D event on INT2. Default value: 0 (0: routing of 6D event on INT2 disabled; 1: routing of 6D event on INT2 enabled)
INT2_TILT	Routing of tilt event on INT2. Default value: 0 (0: routing of tilt event on INT2 disabled; 1: routing of tilt event on INT2 enabled)
INT2_IRON	Routing of soft-iron/hard-iron algorithm end event on INT2. Default value: 0 (0: routing of soft-iron/hard-iron algorithm end event on INT2 disabled; 1: routing of soft-iron/hard-iron algorithm end event on INT2 enabled)

9.83 MASTER_CMD_CODE (60h)

Table 201. MASTER_CMD_CODE register

MASTER_CMD_CODE7	MASTER_CMD_CODE6	MASTER_CMD_CODE5	MASTER_CMD_CODE4	MASTER_CMD_CODE3	MASTER_CMD_CODE2	MASTER_CMD_CODE1	MASTER_CMD_CODE0
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Table 202. MASTER_CMD_CODE register description

MASTER_CMD_CODE[7:0]	Master command code used for stamping for sensor sync. Default: 0
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9.84 SENS_SYNC_SPI_ERROR_CODE (61h)

Table 203. SENS_SYNC_SPI_ERROR_CODE register

ERROR_CODE7	ERROR_CODE6	ERROR_CODE5	ERROR_CODE4	ERROR_CODE3	ERROR_CODE2	ERROR_CODE1	ERROR_CODE0
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Table 204. SENS_SYNC_SPI_ERROR_CODE register description

ERROR_CODE[7:0]	Error code used for sensor synchronization. Default: 0)
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9.85 OUT_MAG_RAW_X_L (66h)

External magnetometer raw data (r).

Table 205. OUT_MAG_RAW_X_L register

D7	D6	D5	D4	D3	D2	D1	D0
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Table 206. OUT_MAG_RAW_X_L register description

D[7:0]	X-axis external magnetometer value (LSbyte)
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9.86 OUT_MAG_RAW_X_H (67h)

External magnetometer raw data (r).

Table 207. OUT_MAG_RAW_X_H register

D15	D14	D13	D12	D11	D10	D9	D8
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Table 208. OUT_MAG_RAW_X_H register description

D[15:8]	X-axis external magnetometer value (MSbyte)
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9.87 OUT_MAG_RAW_Y_L (68h)

External magnetometer raw data (r).

Table 209. OUT_MAG_RAW_Y_L register

D7	D6	D5	D4	D3	D2	D1	D0
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Table 210. OUT_MAG_RAW_Y_L register description

D[7:0]	Y-axis external magnetometer value (LSbyte)
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9.88 OUT_MAG_RAW_Y_H (69h)

External magnetometer raw data (r).

Table 211. OUT_MAG_RAW_Y_H register

D15	D14	D13	D12	D11	D10	D9	D8
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Table 212. OUT_MAG_RAW_Y_H register description

D[15:8]	Y-axis external magnetometer value (MSbyte)
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9.89 OUT_MAG_RAW_Z_L (6Ah)

External magnetometer raw data (r).

Table 213. OUT_MAG_RAW_Z_L register

D7	D6	D5	D4	D3	D2	D1	D0
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Table 214. OUT_MAG_RAW_Z_L register description

D[7:0]	Z-axis external magnetometer value (LSbyte)
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9.90 OUT_MAG_RAW_Z_H (6Bh)

External magnetometer raw data (r).

Table 215. OUT_MAG_RAW_Z_H register

D15	D14	D13	D12	D11	D10	D9	D8
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Table 216. OUT_MAG_RAW_Z_H register description

D[15:8]	Z-axis external magnetometer value (MSbyte)
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9.91 X_OFS_USR (73h)

Accelerometer X-axis user offset correction (r/w). The offset value set in the X_OFS_USR offset register is internally added to the acceleration value measured on the X-axis.

Table 217. X_OFS_USR register

X_OFS_USR_7	X_OFS_USR_6	X_OFS_USR_5	X_OFS_USR_4	X_OFS_USR_3	X_OFS_USR_2	X_OFS_USR_1	X_OFS_USR_0
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Table 218. X_OFS_USR register description

X_OFS_USR_7:0	Accelerometer X-axis user offset correction expressed in two's complement, weight depends on CTRL6_C(4) bit. The value must be in the range [-127 127].
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9.92 Y_OFS_USR (74h)

Accelerometer Y-axis user offset correction (r/w). The offset value set in the Y_OFS_USR offset register is internally added to the acceleration value measured on the Y-axis.

Table 219. Y_OFS_USR register

Y_OFS_USR_7	Y_OFS_USR_6	Y_OFS_USR_5	Y_OFS_USR_4	Y_OFS_USR_3	Y_OFS_USR_2	Y_OFS_USR_1	Y_OFS_USR_0
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Table 220. Y_OFS_USR register description

Y_OFS_USR_7:0	Accelerometer Y-axis user offset correction expressed in two's complement, weight depends on CTRL6_C(4) bit. The value must be in the range [-127 127].
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9.93 Z_OFS_USR (75h)

Accelerometer Z-axis user offset correction (r/w). The offset value set in the Z_OFS_USR offset register is internally subtracted from the acceleration value measured on the Z-axis.

Table 221. Z_OFS_USR register

Z_OFS_USR_7	Z_OFS_USR_6	Z_OFS_USR_5	Z_OFS_USR_4	Z_OFS_USR_3	Z_OFS_USR_2	Z_OFS_USR_1	Z_OFS_USR_0
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Table 222. Z_OFS_USR register description

Z_OFS_USR_7:0	Accelerometer Z-axis user offset correction expressed in two's complement, weight depends on CTRL6_C(4) bit. The value must be in the range [-127 127].
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